

ALPHA MODEL WITH

CNN-LSTM-Attention

Professional Backtesting & Alpha Signal Generation on S&P 500

ELENE 4904 — Statistical Learning in Quantitative Trading
April 28, 2026

AGENDA

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Project Overview & Motivation

The Alpha Generation Problem

- Markets are informationally competitive — pure price prediction is near-impossible
- Cross-sectional alpha: rank stocks by expected relative return, not absolute
- Classic factors (momentum, value) have decayed — ML can discover non-linear combinations
- Challenge: overfitting, lookahead bias, transaction costs erode edge

What We Built

- 496 S&P 500 stocks, 10+ years of data
- 24 engineered alpha features
- CNN-LSTM-Attention neural network
- Full long-short portfolio backtest
- Risk budgeting + transaction cost model
- Benchmarks: Ridge, XGBoost, Momentum, Mean Reversion

496

Stocks in Universe

24

Alpha Features

12

Years of Data

~1.37M

Training Sequences

24 features in 6 groups

1

Returns

ret_1/5/10/21/63 d
excess_ret_1/5d vs SPY
ret_5/21/63d cross-rank

Captures momentum across horizons

2

Volatility

vol_5/21/63 d
vol_21d cross-z + percentile
beta_63 vs SPY

Risk regime + market sensitivity

3

Price vs SMA

price_to_sma_10/21/63
cross-sectional z-score
daily_range, close_position

Where is price within recent band

4

Volume / Liquidity

volume_chg_1d
volume_to_sma_21
dollar_volume_to_sma_21

Conviction behind a move

5

Microstructure

close_open_ret
daily_range
cross-stock dispersion

Intraday behavior signals

6

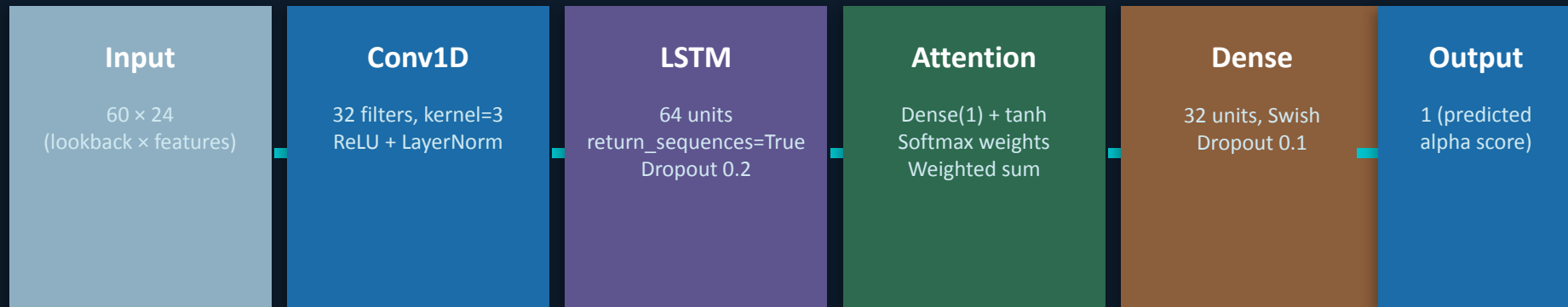
Cross-section

z-score per date
percentile rank
(applied to 8 base features)

Removes day-by-day market drift

Target: 5-day forward return. Lag: 1 day execution lag. Normalization: StandardScaler fit on train only + cross-sectional z per date.

Model Architecture: CNN-LSTM-Attention



Design Rationale

- ◆ Conv1D captures local short-term patterns in the 60-day sequence (momentum pulses, volatility regimes)
- ◆ LSTM retains long-range temporal dependencies — remembers context from weeks ago
- ◆ Attention mechanism weights each time step differently, highlighting the most predictive days in the lookback window
- ◆ Huber loss (δ -insensitive) reduces sensitivity to outlier return observations common in equity data
- ◆ Adam optimizer ($\text{lr}=0.0005$) + EarlyStopping (patience=5) · trained to convergence

LSTM & CNN-LSTM Theory — Why It Fits This Project

What an LSTM Adds

Memory cell: keeps useful information across many trading days

Input / forget gates: decide what new signals to store and what stale signals to drop

Output gate: translates the hidden state at each time step; these are later translated into output scores



Why CNN + LSTM + Attention?



Conv1D: detects local motifs: price bursts, volatility spikes, reversal patterns



LSTM: models how those motifs persist or fade through the 60-day window



Attention: lets the model focus on the most relevant days instead of treating all days equally

Fit for this project: equity alpha is sequential, noisy, non-linear, and regime-dependent — exactly the setting where sequence models can improve on one-day tabular regressions.

Training Setup & Baseline Models

Train

Jan 2014 – Dec 2020

773,072 sequences

Validation

Jan 2021 – Dec 2022

218,521 sequences

Test (OOS)

Jan 2023 – Apr 2026

379,241 sequences

Baseline Models Compared

Ridge Regression

$\alpha=10$

Tabular (last timestep only) · linear baseline

XGBoost / HistGBR

$n=400$

Gradient boosted trees on tabular features

Momentum

21d ret

Buy past winners — classical factor

Mean Reversion

-5d ret

Buy past losers — contrarian factor

CNN-LSTM Training Details

- Epochs: trained with EarlyStopping (patience 5) on a held-out 2021-2022 validation block
- Batch size: 512 · Optimizer: Adam (lr=0.0005)
- Loss: Huber
- Input shape: (batch, 60 timesteps, 24 features)

Rank IC Diagnostics — Signal Quality

Rank IC = Spearman correlation between predicted and realized 5-day returns, computed cross-sectionally each day

Metric	CNN-LSTM-Attn	Ridge	XGBoost	Momentum	Mean Rev.
Mean Rank IC	0.0362 ★	0.0109	0.0223	-0.0004	0.0006
IC Std	0.1705	0.1812	0.1537	0.1792	0.1683
IC t-stat	5.4647 ★	1.66	4.00	-0.06	0.10
Positive IC %	56.99% ★	51.24%	54.12%	51.64%	47.39%
IC Count	765	765	765	825	825

Key Takeaways

CNN-LSTM has the strongest ranking signal in the test period: IC t-stat 5.46

Ridge is marginal; XGBoost is solid, indicating non-linear structure in the features

Momentum and mean reversion are near zero in this setup

Important: Rank IC measures cross-sectional signal quality, not pure portfolio alpha

Backtesting Framework

1

Signal Generation

Model predicts 5-day alpha score for every stock each rebalance day

2

Stock Selection

Long: top 10% by predicted score
Short: bottom 10% by predicted score

3

Risk-Weighted Sizing

Weight = $1 / \text{vol_21d}$
Cap at MAX_POSITION = 8% per stock

4

Normalization

Iterative capping-and-renormalization
Dollar-balanced, but beta still monitored

5

Transaction Costs

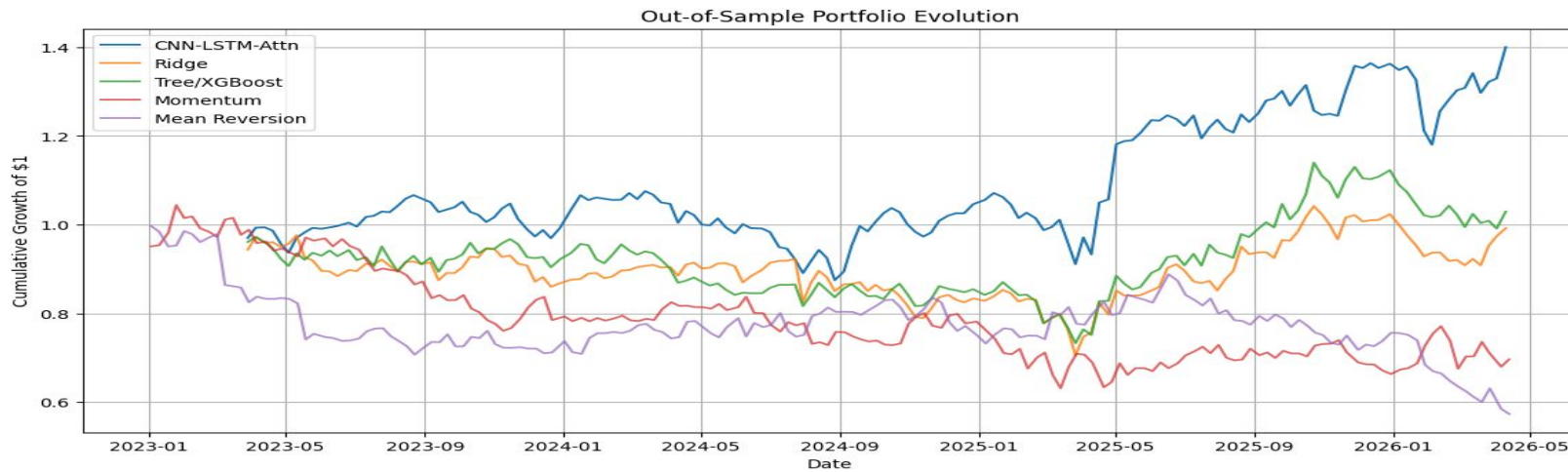
0.1% one-way on all weight changes
Rebalance every 5 trading days

6

P&L Attribution

Per stock contribution tracked
Sector, beta, long/short decomposition

Out-of-Sample Portfolio Evolution (2023–2026)



+40.02%

Total Return (CNN-LSTM)

11.73%

Annualized Return

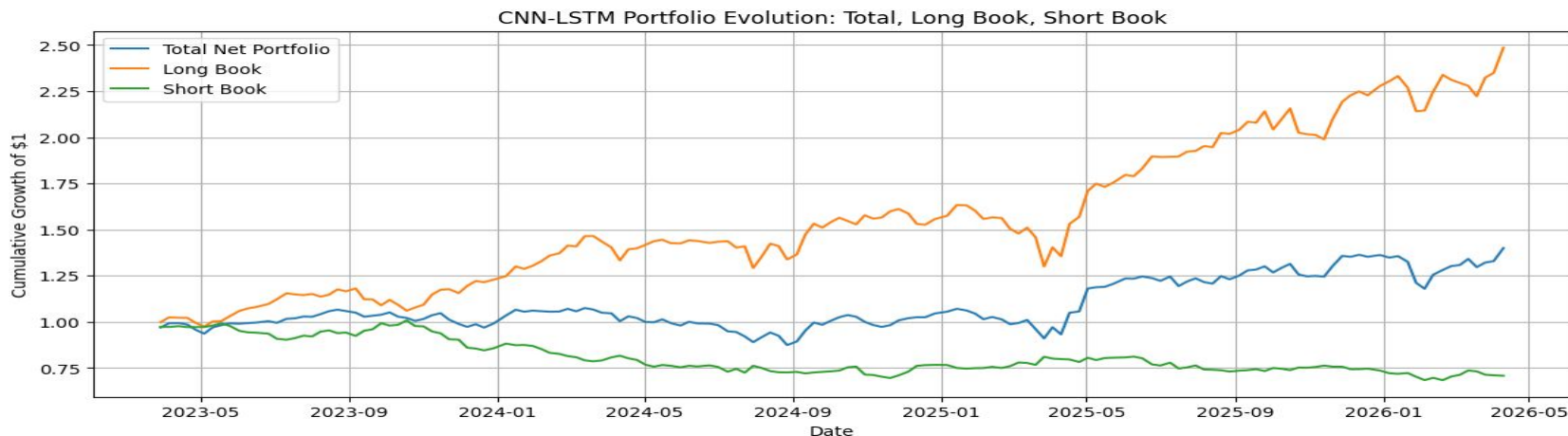
0.67

Sharpe Ratio

+40 PP

Outperformance vs Baselines

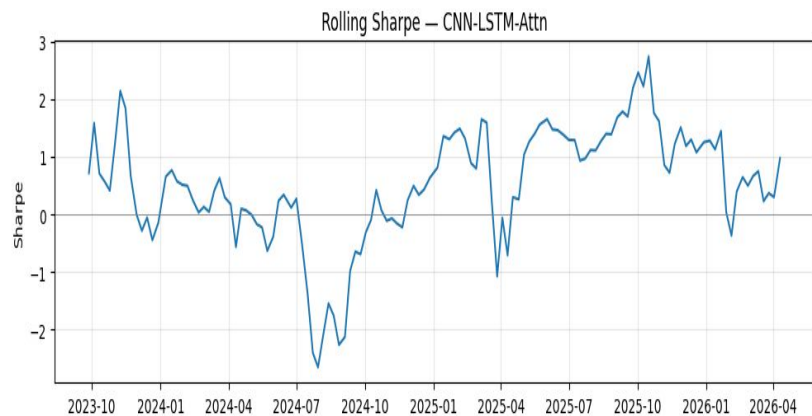
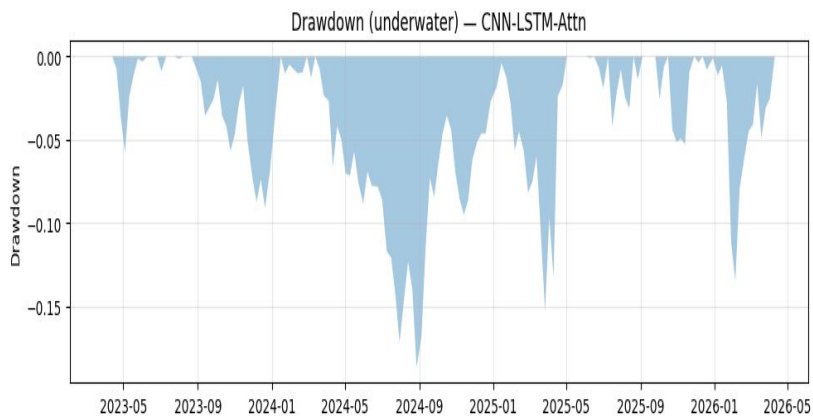
Long / Short Book Decomposition



Interpretation

Long book is the main return driver, especially during the 2025 growth/technology rally. The short book is less consistent, so the result should be interpreted as a strong ranking signal with room for better beta and sector control.

Drawdown & Rolling Sharpe Ratio



-18.66%

Max Drawdown

248 days

Recovery Period

0.6284

Calmar Ratio

1.1654

Sortino Ratio

Comprehensive Performance Metrics

§ Period Metrics

Metric	Value
Total Return	40.02%
Ann. Return	11.73%
Ann. Volatility	19.33%
Sharpe Ratio	0.67
Sortino Ratio	1.17
Max Drawdown	-18.66%
Calmar Ratio	0.63
Skewness	0.89
Excess Kurtosis	4.39
Ulcer Index	0.059

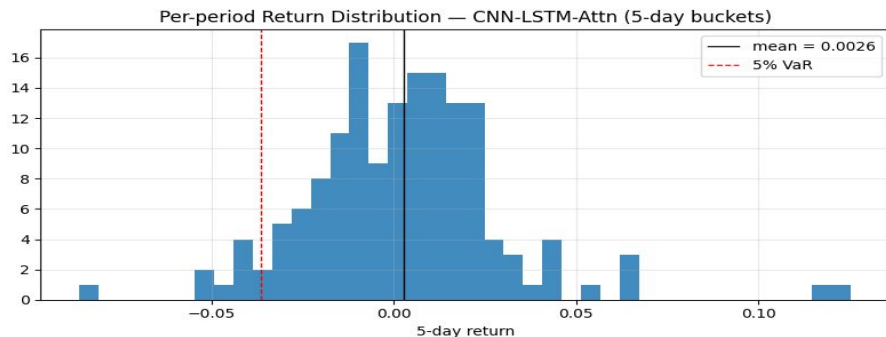
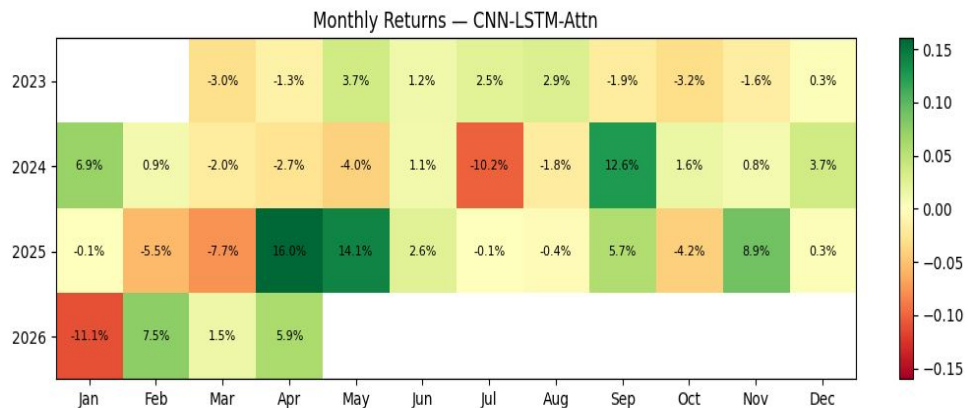
§ Trade Metrics

Metric	Value
# Trades	3,478
Win Rate	54.89%
Avg Win	+0.163%
Avg Loss	-0.156%
Win / Loss	1.045x
Profit Factor	1.2712
Avg Hold (days)	29.8
Avg Turnover	1.7682

§ Risk / Exposure

Metric	Value
Gross Exposure	2.0x
Net Exposure	≈ 0
Net Beta	0.5190
HHI Conc.	0.021
Top-5 Share	19.8%
Ann. Alpha	13.2%
Track. Error	23.36%
VaR 95%	-3.65%
CVaR 95%	-5.10%

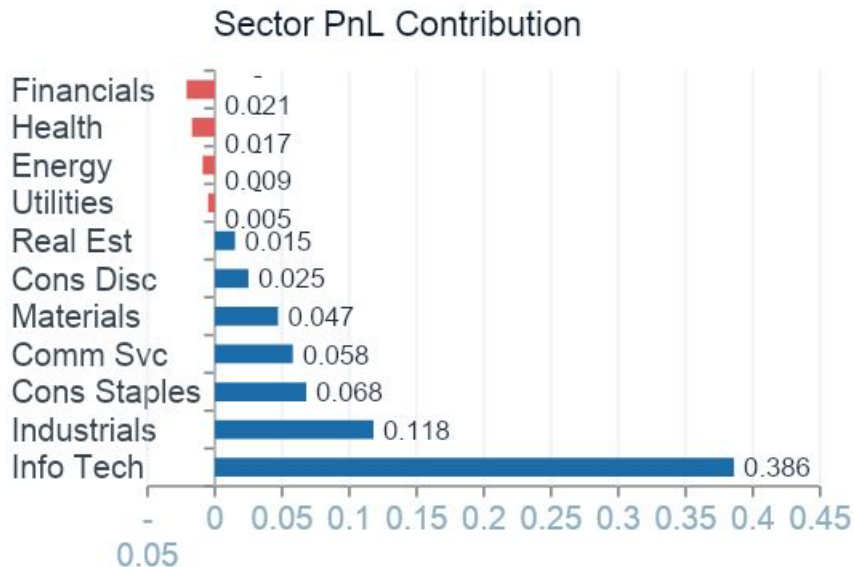
Monthly Returns Heatmap & Distribution



Observations

2025 was the strongest year
— Sharpe ≈ 1.28
Jan 2026 drawdown was quickly followed by recovery
Distribution is positively skewed but fat-tailed
Most losses cluster near the 5% VaR line

Sector Attribution & Top Ticker P&L

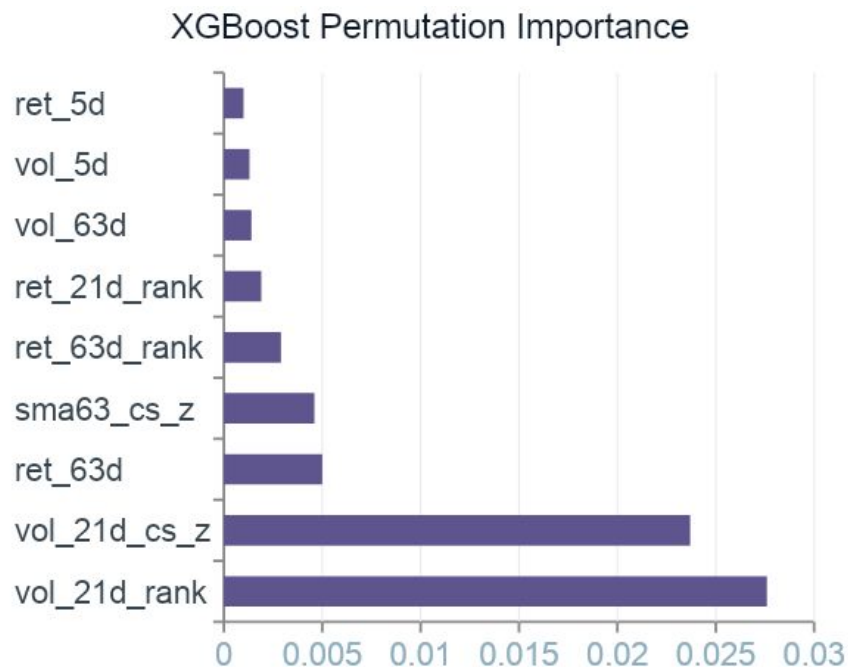
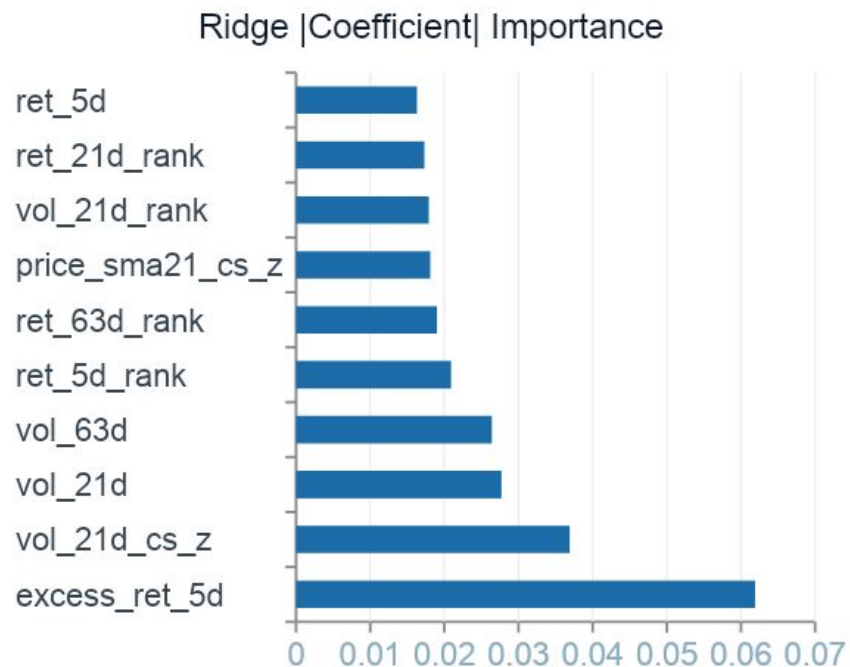


Most Held Stocks

Ticker	Times Held	Cum. PnL
CVNA	137	+1.95%
PLTR	128	+1.65%
APP	124	+1.98%
FIX	106	+4.21%
FSLR	93	+2.80%
MU	80	+3.31%
MRNA	84	-2.71%
FISV	—	-3.29%

Key Insight: Information Technology is the largest P&L contributor, but this also reveals sector concentration. A production version should add sector-neutral constraints before claiming pure alpha.

Feature Importance Analysis



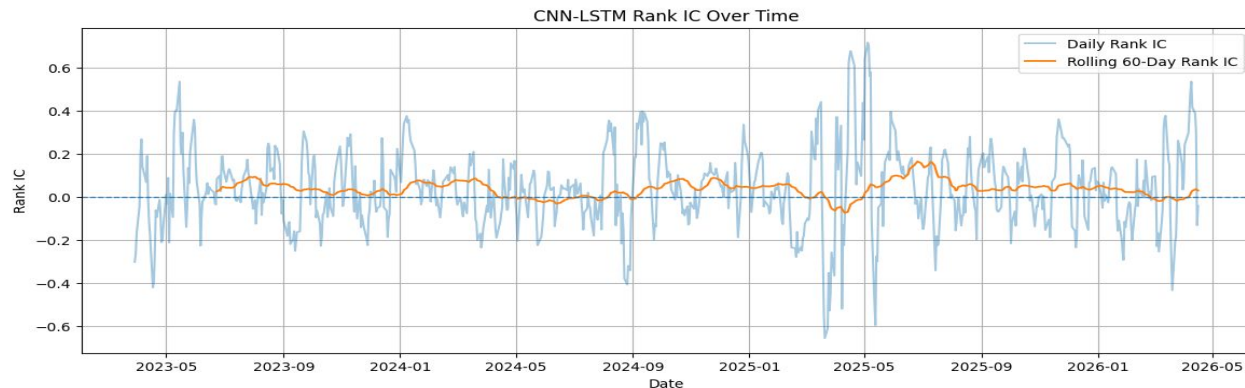
Takeaway: Volatility, excess-return, and cross-sectional momentum variables carry most of the signal in the baselines. This supports the idea that the neural model is learning non-linear versions of economically meaningful factors.

Yearly Performance & Rank IC Over Time

Annualized portfolio performance by year

Year	Ann. Return	Ann. Vol	Sharpe	Max DD	Hit Rate
2023	-0.96%	12.0%	-0.08	-9.1%	56.4%
2024	+5.44%	16.6%	0.33	-18.7%	44.0%
2025	+30.5%	23.9%	1.28	-15.0%	58.0%
2026 YTD	+10.3%	26.8%	0.39	-13.0%	64.3%

Rank IC over time: signal quality is persistent but noisy



Takeaway

- 2025 is the strongest year.
- Positive OOS result is not uniform over time.
- Rank IC confirms a real ranking signal, but it still needs risk controls.

Concrete upgrades for the next iteration

Direction	Why	Implementation
Market-residual target	Predict residual return after regressing each stock on SPY, so the model learns pure idiosyncratic alpha rather than re-discovering market direction.	Re-define y as residual; add macro features (VIX, yield-curve, credit spreads).
Explicit beta neutralization	Add an active β_{63} constraint to the portfolio optimizer so position-weighted forward beta is held tight to zero, complementing realized beta neutrality.	$ \text{net } \beta_{63} \leq 0.10$ hard constraint; optional SPY-overlay hedge sleeve.
Sector-neutral construction	Balance gross exposure across GICS sectors so no single sector (e.g. Information Tech) dominates the book and the strategy diversifies idiosyncratic risk.	Per-sector long/short caps; equal gross weight per GICS sector.
Tail-risk-aware loss	Penalize the model for fat-tail behavior during training so VaR/CVaR estimates better track realized extreme losses.	CVaR-adjusted training loss; vol-regime circuit-breakers in deployment.
Richer feature set	Bring in macro state, volatility-surface signals, and intraday microstructure to widen the feature universe beyond the 24 price-derived features used here.	Add VIX term-structure, IV skew, sector ETF momentum, news/sentiment scores.

CNN-LSTM-Attention Finds a Strong OOS Ranking Signal

- ✓ IC t-stat of 5.46 shows statistically meaningful cross-sectional ranking power
- ✓ 40% total return, 12% annualized, Sharpe 0.67 over the OOS period
- ✓ Portfolio result is positive, but not yet cleanly beta-neutral or sector-neutral
- ✓ CNN-LSTM outperforms simple factor baselines in this experimental setup
- ✓ Main next step: market-residual target + beta/sector constraints + stronger validation